VICTORY 2027

LIGHT

LECTURE-03 PHYSICS

Shailendra sir

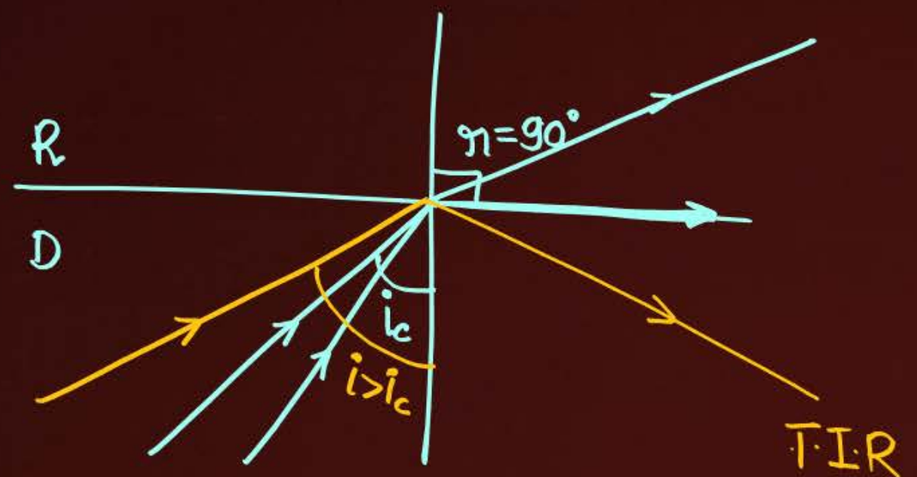


Topics To Be Covered

1. REVISION
2. TOTAL INTERNAL REFLECTION IN A PRISM
3. TOTAL INTERNAL REFLECTION IN A PRISM
4. QUESTIONS


SHAILENDRA SIR
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REVISION



(i) $D \rightarrow R$

(ii) $i > i_c$

For Air-Glass Interface,

$$\sin i_c = \frac{1}{{}_a\mu_g}$$

$$= \frac{1}{\left(\frac{3}{2}\right)}$$

$$\sin i_c = \frac{2}{3}$$

$$\boxed{i_c = \sin^{-1}\left(\frac{2}{3}\right) \approx 42^\circ}$$

TOTAL INTERNAL REFLECTION IN A PRISM

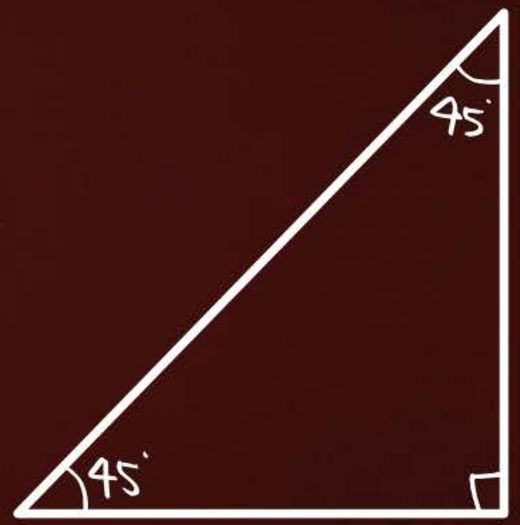
T.I.R. THROUGH A $45°, 90°, 45°$ PRISM (OR RIGHT-ANGLED ISOSCELES PRISM)

Total reflecting prism: A prism having an angle of $90°$ between its two refracting surfaces and the other two angles each equal to $45°$, is called a total reflecting prism because the light incident normally on any of its faces, suffers total internal reflection inside the prism. Due to this behaviour, a total reflecting prism is used for the following three purposes :

(a) to deviate a ray of light through $90°$,

(b) to deviate a ray of light through $180°$, and

(c) to erect the inverted image without producing deviation in its path.



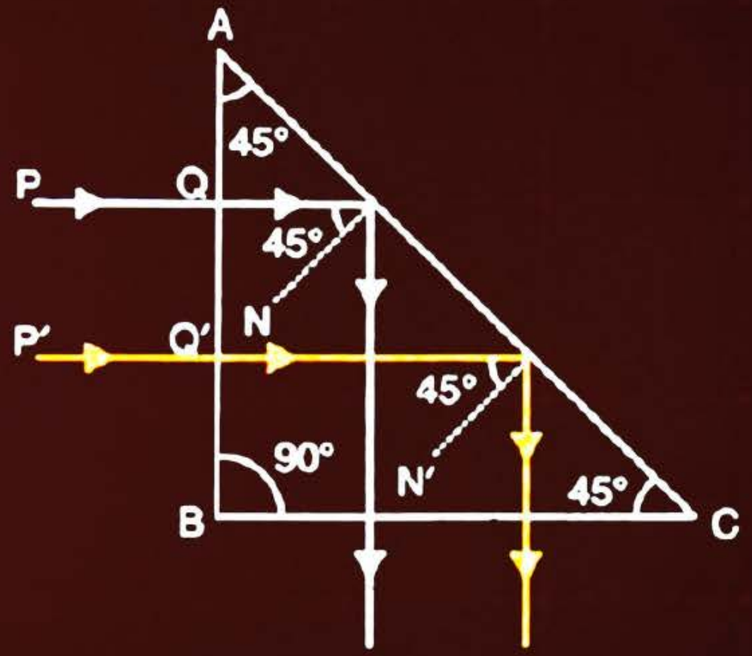
TOTAL INTERNAL REFLECTION IN A PRISM

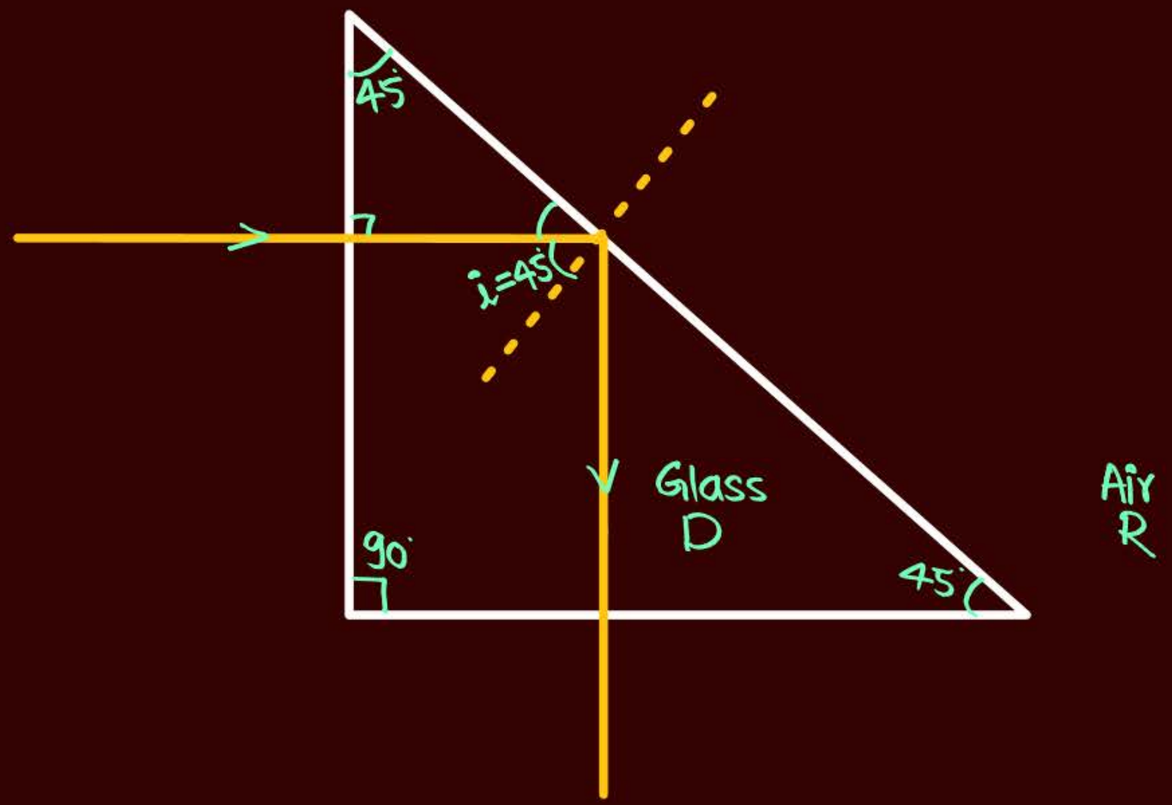
T.I.R. THROUGH A $45°, 90°, 45°$ PRISM (OR RIGHT-ANGLED ISOSCELES PRISM)

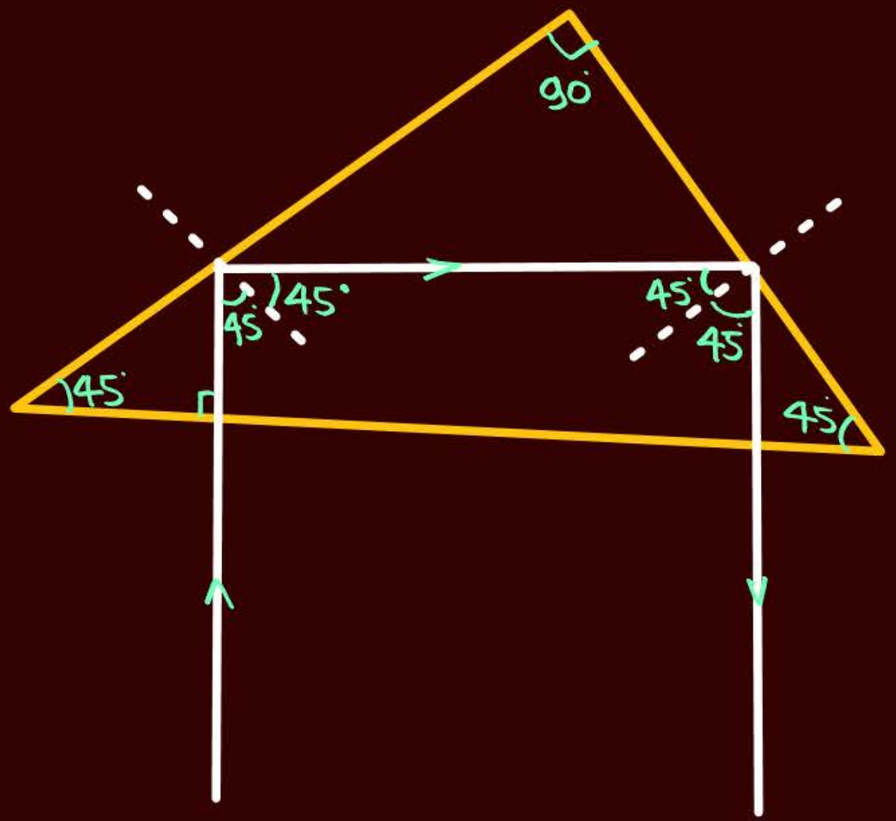
TO DEVIATE A RAY OF LIGHT THROUGH $90°$

For glass-air interface, the critical angle is about $42°$, therefore the beam of light suffers total internal reflection at face AC because the angle of incidence is greater than the critical angle.

- By the principle of reversibility of path of light, if light is incident normally on the face BC of the prism, it will emerge out from the face AB after suffering total internal reflection at the face AC .
- Use: This action of prism is used in a periscope where a total reflecting prism is preferred over a plane mirror.




45°
i=45°
Glass
D
Air
R
90°
45°


90°
45°
45°
45°
45°
45°
45°

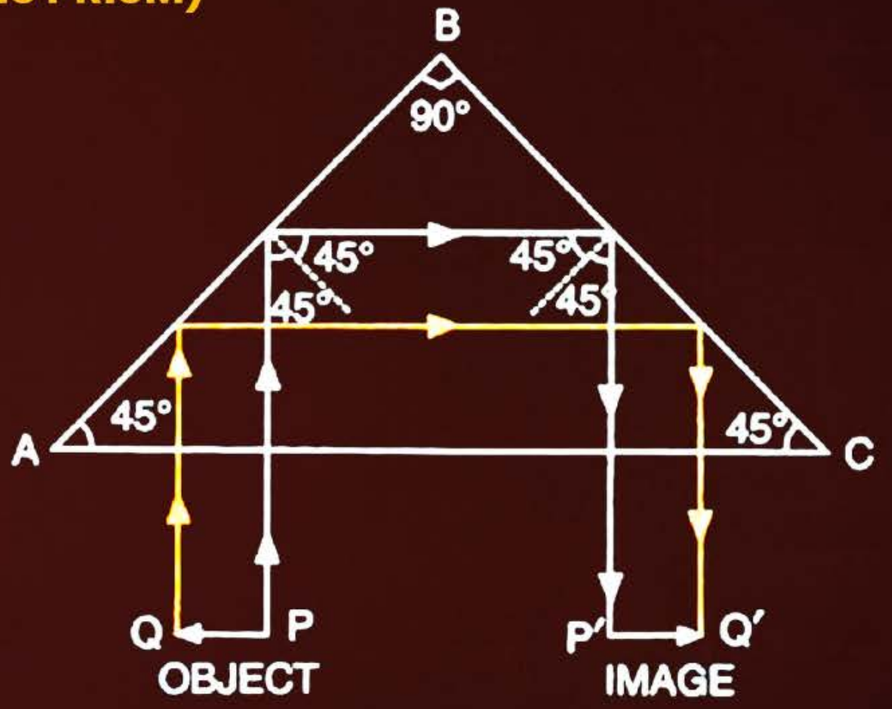
TOTAL INTERNAL REFLECTION IN A PRISM

T.I.R. THROUGH A $45°, 90°, 45°$ PRISM (OR RIGHT-ANGLED ISOSCELES PRISM)

TO DEVIATE A RAY OF LIGHT THROUGH $180°$

Since the beam of light is travelling from glass to air at an angle of incidence equal to $45°$ which is greater than the critical angle ($= 42°$ for glass-air), therefore the beam of light suffers total internal reflection. The reflected beam inside the prism strikes the face BC at an angle of incidence equal to $45°$, so it again suffers total internal reflection.

- The beam of light inside the prism now falls normally on the face AC and therefore passes undeviated out of the prism, forming the image $P'Q'$ of the object PQ.
- The beam thus gets deviated by $90°$ at each reflection and the total deviation due to two reflections becomes $180°$.

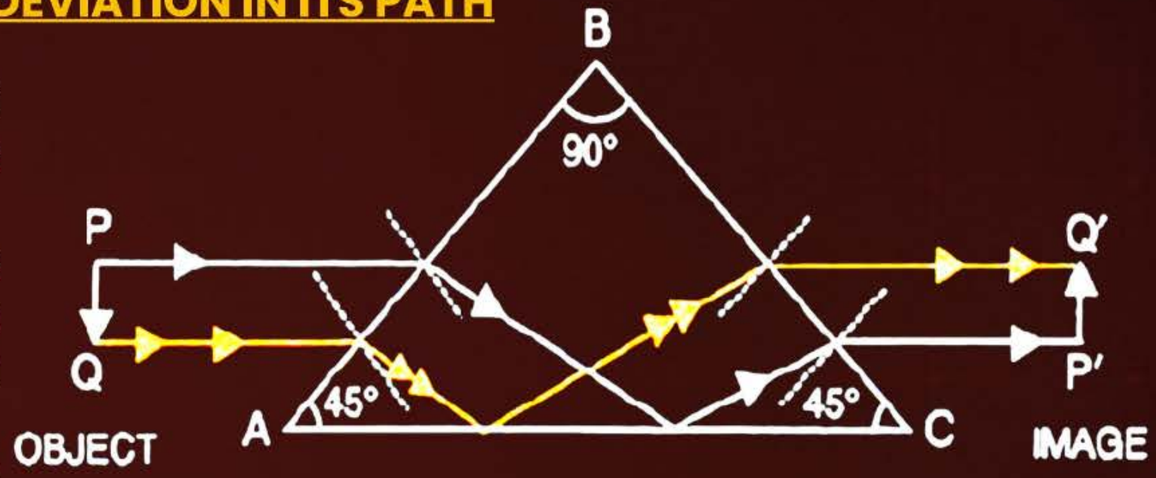


Use : This action of prism is **used in a binocular and camera** to invert the image without the loss of intensity.

TOTAL INTERNAL REFLECTION IN A PRISM

T.I.R. THROUGH A $45°, 90°, 45°$ PRISM (OR RIGHT-ANGLED ISOSCELES PRISM)

TO ERECT THE INVERTED IMAGE WITHOUT PRODUCING DEVIATION IN ITS PATH



- The beam of light from the object PQ is incident parallel to the face AC of the prism and strikes the face AB of the prism.
- It suffers refraction from air to glass and strikes the face AC of the prism travelling from glass to air at an angle of incidence greater than the critical angle(=42), therefore it suffers total internal reflection.
- The beam inside the prism now strikes the face BC at an angle of incidence less than the critical angle

hence it suffers refraction from glass to air and bends away from the normal. The beam emerges parallel to the face AC . As a result of refraction, on emergence the rays are inverted and for the inverted object PQ , the erect image $P'Q'$ is obtained. A prism when used in this manner is called the erecting prism.

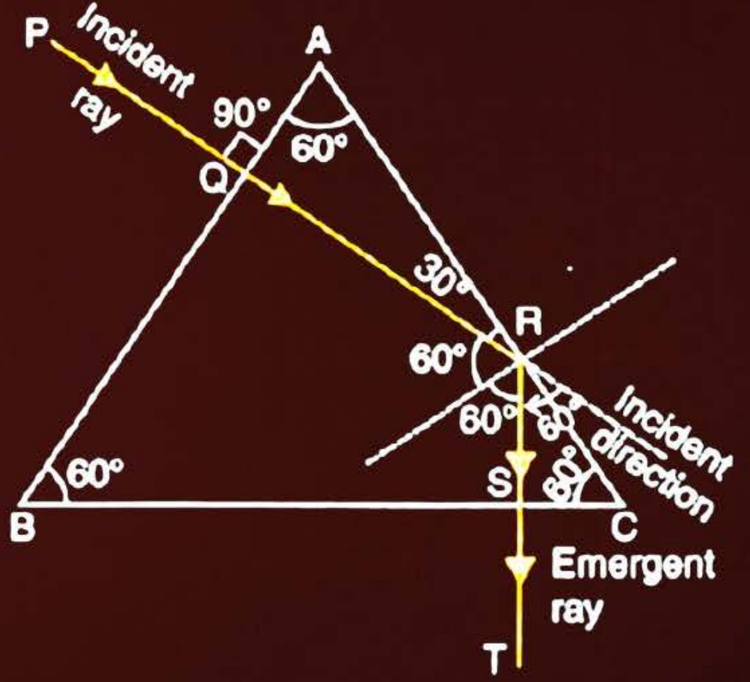
Use: This action of prism is used in a slide projector.

TOTAL INTERNAL REFLECTION IN A PRISM

T.I.R. THROUGH PRISM WHERE EACH ANGLE IS $60°$ (EQUILATERAL PRISM)

A light ray PQ incident normally on the face AB of the prism passes undeviated as QR into the prism and it strikes the face AC at an angle of incidence $60°$ at the point R.

- Now the light ray QR suffers total internal reflection at the glass-air interface, since the angle of incidence $(= 60°)$ is greater than the critical angle which is $42°$, such that $\angle QRS$ is $120°$.
- The ray RS obtained after total internal reflection falls normally on the face BC of the prism and so it passes out undeviated through the face BC. Thus ST is the emergent ray.
- From Fig., it is clear that the incident ray PQ has turned through an angle of 60 towards the base from its initial direction and it emerges out as ST.

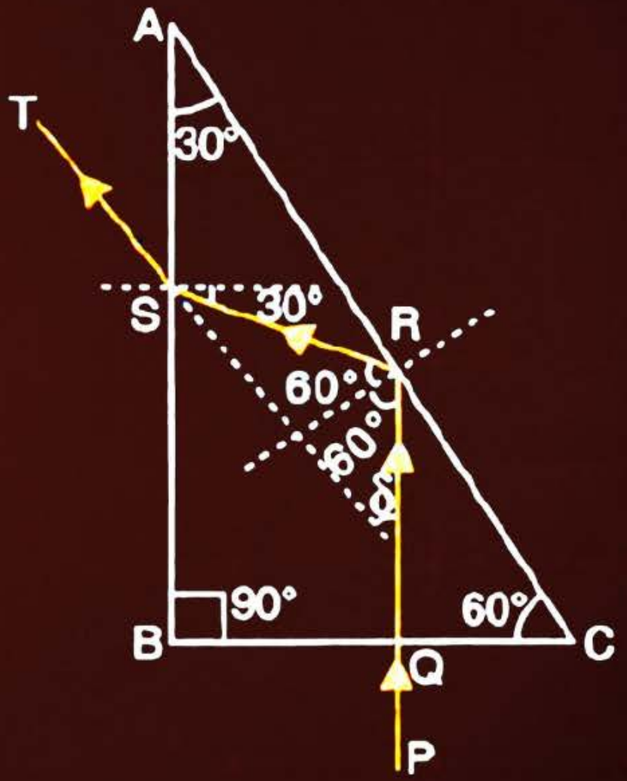


TOTAL INTERNAL REFLECTION IN A PRISM

T.I.R. and refraction of light through $30°, 90°, 60°$ prism (right angled prism)

A $30°, 90°, 60°$ prism can be used to deviate a light ray through an angle less than $60°$ by total internal reflection.

- A light ray PQ incident normally on the face BC of the prism (opposite to the $30°$ refracting angle) passes undeviated as QR into the prism and it strikes the face AC at an angle of incidence $60°$ at the point R.
- Now the light ray QR suffers total internal reflection at the glass-air interface since the angle of incidence $(= 60°)$ is greater than the critical angle which is $42°$.
- The ray RS obtained after total internal reflection is incident at the face AB of the prism at an angle of $30°$.
- This angle of incidence at glass-air interface is less than the critical angle, so the ray RS is refracted from glass to air as ST with an angle of refraction greater than $30°$.
- Thus the emergent ray ST has turned through an angle δ (which is less than $60°$) from the direction of incident ray PQ.

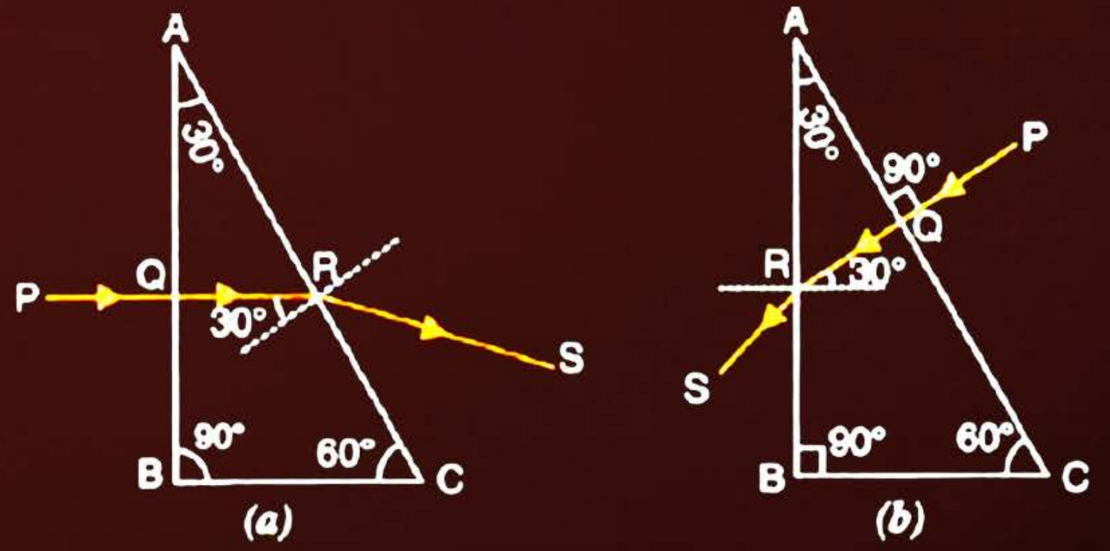


TOTAL INTERNAL REFLECTION IN A PRISM

NON OCCURANCE OF T.I.R. THROUGH A $30°, 90°, 60°$ PRISM

If a light ray PQ is incident normally on the face AB or a part of face AC of the prism ABC , it does not suffer total internal reflection inside the prism and gets refracted as RS obeying the laws of refraction as shown in Fig. (a) and (b).

- In each case, the angle of incidence at the other face inside the prism is $30°$ which is less than the critical angle (i.e., $42°$).

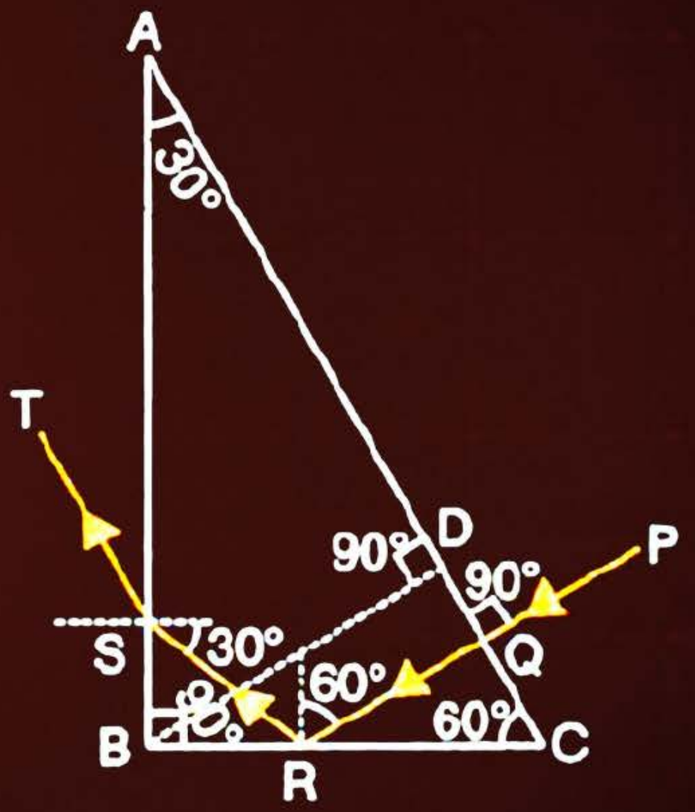


TOTAL INTERNAL REFLECTION IN A PRISM

EXCEPTION

A ray of light incident normally on hypotenuse of prism below the foot of perpendicular on it from its opposite corner, suffers total internal reflection and gets deviated by an angle greater than $60°$.

- In Fig., a light ray PQ is incident normally on the face AC (hypotenuse) of the prism in the portion DC such that the point D is the foot of perpendicular from B on AC.
- It passes undeviated as QR inside the prism and strikes the face BC of the prism at an angle of incidence $60°$ which is greater than the critical angle $(= 42°)$. The ray suffers total internal reflection inside the prism as RS.
- The reflected ray RS then strikes the face AB at an angle of incidence $30°$, so it suffers refraction from glass to air and emerges out of the prism as ST, bending away from the normal at the face AB.
- Thus the incident ray PQ gets deviated through an angle greater than $60°$.



USE OF A TOTAL INTERNAL REFLECTING PRISM IN PLACE OF A PLANE MIRROR

- A total reflecting prism is used in place of a plane mirror to deviate the light ray by $90°$ in a periscope and by $180°$ in a binocular as well as in a camera.
- The reason is that due to total internal reflection in the prism, the entire incident light (100%) is reflected back into the denser medium, whereas in ordinary reflection from a plane mirror, some light is refracted and absorbed, so the reflection is not 100%.
- Thus the image obtained by the use of a total reflecting prism, is much brighter than that obtained by using a plane mirror.
- Further the brightness of image formed by a total reflecting prism always remains unchanged, while due to deterioration of silvering of the plane mirror after a long use, the image formed by mirror becomes faint.

DISTINCTION BETWEEN THE TOTAL INTERNAL REFLECTION AND REFLECTION FROM A PLANE MIRROR

HW

	Total internal reflection		Reflection from a plane mirror
1.	It takes place only when light passes from a denser medium to a rarer medium at an angle of incidence greater than the critical angle for that pair of media.	**1.**	It takes place when light is incident on a plane mirror from any medium at any angle of incidence.
2.	The entire light is reflected.	**2.**	Only a part of light is reflected while the rest is refracted and absorbed.

DISTINCTION BETWEEN THE TOTAL INTERNAL REFLECTION AND REFLECTION FROM A PLANE MIRROR

	Total internal reflection		Reflection from a plane mirror
3.	There is no loss of energy. The energy of the reflected ray is the same as that of the incident ray.	**3.**	There is a loss of energy. The energy of the reflected ray is always less than that of the incident ray.
4.	The image is much brighter, and the brightness remains unchanged even after long use of the total reflecting device.	**4.**	The image is less bright, and the brightness gradually decreases as the silvering on the mirror deteriorates with long use.

SOME CONSEQUENCES OF TOTAL INTERNAL REFLECTION

(1) On a hot sunny day, a driver may see a pool of water (or wet road) in front of him at some distance. It is the phenomenon of mirage which is often observed in desert.

(2) An empty test tube placed in water in a beaker with mouth outside the water surface, shines like a mirror when seen at certain angles.

(3) A crack in a glass vessel often shines like a mirror.

(4) A piece of diamond sparkles when viewed from certain directions.

(5) An optical fibre is used to transmit a light signal over a long distance without the loss of energy. However, some energy gets lost due to absorption of light in its material.

QUESTIONS

The critical angle for glass-air interface is:

1. 24°
2. 48°
3. 42°
4. 45°

QUESTIONS

A total reflecting right angled isosceles prism can be used to deviate a ray of light through:

1. 30°
2. 60°
3. 75°
4. 90°

QUESTIONS

A total reflecting equilateral prism can be used to deviate a ray of light through:

1. 30°
2. 60°
3. 75°
4. 90°

QUESTIONS

An endoscope uses optical fibre to transmit high resolution images of internal organs without loss of information. The phenomenon of light that governs the functioning of the optical fiber is:

[ICSE SQP 2024]

1. refraction
2. reflection
3. scattering
4. total internal reflection.

QUESTIONS

Diamonds glitter in the dark because:

1. they emit light.
2. they have a very small critical angle due to a very high refractive index.
3. of the fluorescence.
4. chemical reaction in the diamond produces light energy.

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You